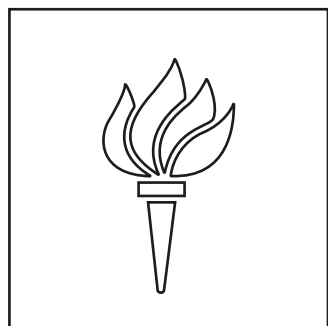


NYU Ability Project

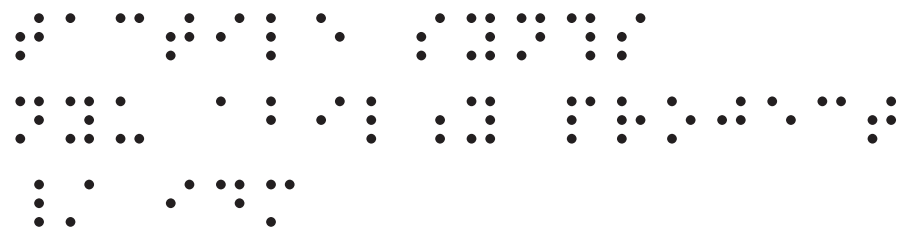
Serge Modular 73-75 Panels



NYU

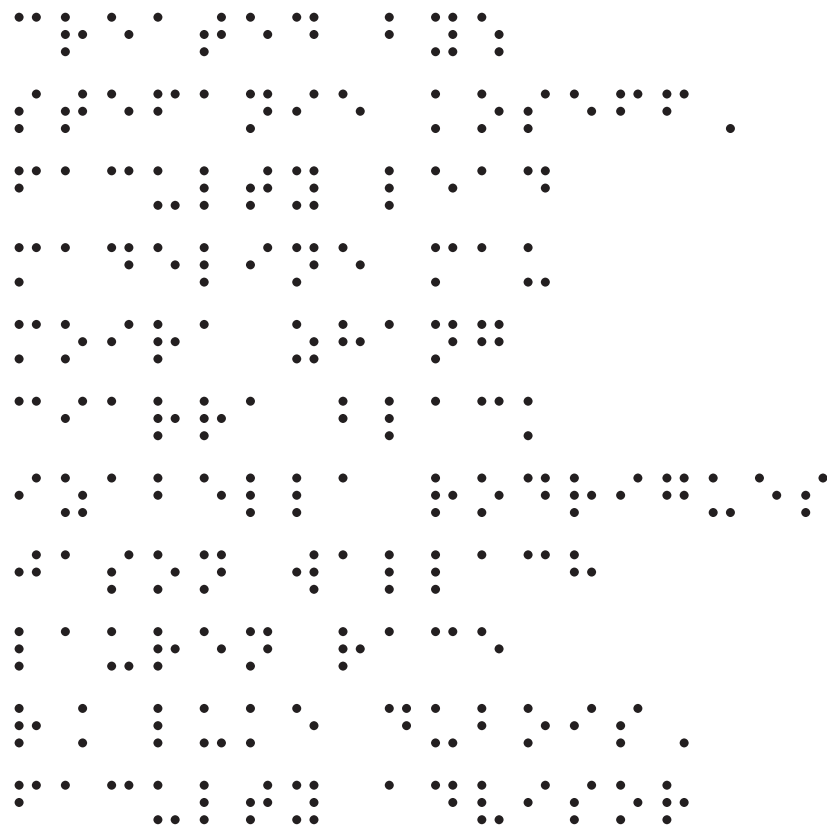
Ability
Project

NYU Ability Project



Tactile Synths

NYU Ability Project / IDM



Created by:

Stefanie Koseff, Faculty Lead

Madeline Mau

Moirra Zhang

Ciarra Black

Izabella Rodrigues

Jason Wallach

Lauren Race

R. Luke DuBois, Faculty Advisor

panel guide

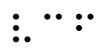
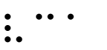
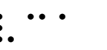
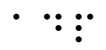
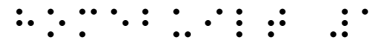
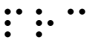
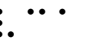
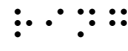
 VOICE	 OSC	 OSC	 TWS	 VCF	 VCA	 VCA	 RING	 ENV	 ADP
 CONTROL	 NOI	 POS	 NEG	 COM	 PRC	 SSG			
 HOMEBUILT #1	 OSC	 TWS	 P&T	 COM	 PRC	 VCA	 RING	 RVB	 PRE
 HOMEBUILT #2	 POS	 NEG	 ENV	 ENV	 ENV	 ADP			
 PRESET	 SEQ	 PRG	 TBS						

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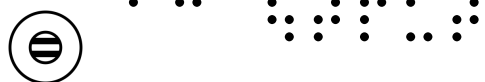
KEY	1	PRC	6	POS	11	VCF	16
OSC	2	VCA	7	NEG	12	SSG	17
TWS	3	RING	8	ENV	13	SEQ	18
P&T	4	RVB	9	ADP	14	PRG	19
COM	5	PRE	10	NOI	15	TBS	22

KEY

symbol key



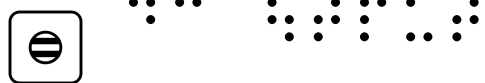
ac input



ac output



dc input



dc output



pulse input



pulse output



misc input



misc output



basic knob



offset knob



scale knob



button



trs input



trs output

OSC

oscillator



· · · · ·

sine ac



· · · · ·

sine dc



· · · · ·

saw dc



· · · · ·

fm



· · · · ·

fm x



· · · · ·

sync



· · · · ·

ws x



· · · · ·

waveshape



· · · · ·

saw ac



· · · · ·

fm



· · · · ·

fm x



· · · · ·

frequency

TWS

triple waveshaper

   a vc1	   a vc2	   a out
   a ×	   a in	   a out
   b vc1	   b vc2	   b out
   b ×	   b in	   b out
   c vc1	   c vc2	   c out
   c ×	   c in	   c out

⠠⠏⠠⠞ P&T

⠠⠏⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak & trough

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak out

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak out

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ peak in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ trough out

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ trough out

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ trough in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ trough in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ trough in

⠠⠞⠠⠞⠠⠞⠠⠞ ⠠⠞⠠⠞⠠⠞⠠⠞ trough in

triple comparator

	   		   
a in 1		a out	
	     		     
a thresh		a in 2	
	   		   
b in 1		b out	
	     		     
b thresh		b in 2	
	   		   
c in 1		c out	
	     		     
c thresh		c in 2	

PRC

dual processor

					
a in 1		a 1 ×		a out	
					
a in 2		a 2 ×		a out	
					
a in 3		a 3 ×		a offset	
					
b in 1		b 1 ×		b out	
					
b in 2		b 2 ×		b out	
					
b in 3		b 3 ×		b offset	

VCA gate

	
out	
	
ac	
	
dc	
	
lin	
	
log	
	
gain	



out



y



VC y



X



VC X


$$X - X$$

RVB

reverbulator



out



out



mix



wet

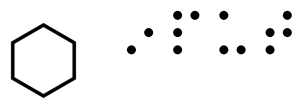


wet



in

preamplifier



input



gain



out



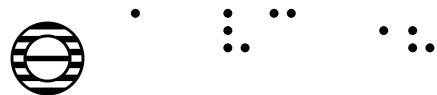
out

POS

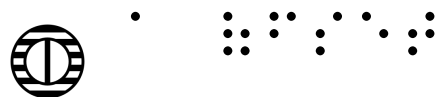
positive slew



a vc



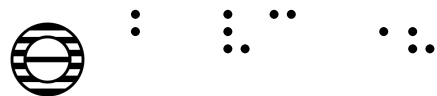
a vc x



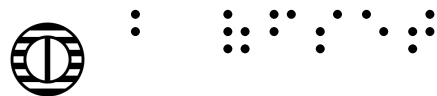
a offset



b vc



b vc x



b offset



a in



a start



a sustain



b in



b start



b sustain



a out



a high



a end



b out



b high



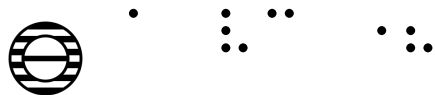
b end

⠠⠠⠠⠠⠠ NEG

⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ negative slew



a vc



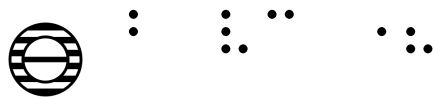
a vc x



a offset



b vc



b vc x



b offset



a in



a end



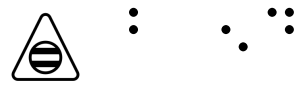
a end



b in



b end



b end



a out



a out



b out



b out

ENV

envelope generator



end



start in



cycle



rise



out



out



hold



fall



window



window offset



window vc



duration vc



duration ×

ADP adapter

	 
a out	
	 
a in	
	 
b out	
	 
b in	
	 
c out	
	 
c in	

NOI

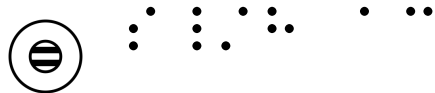
noise source



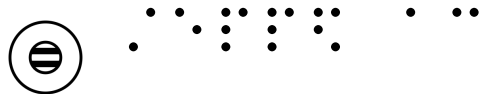
white



pink



s/h ac



stepped ac



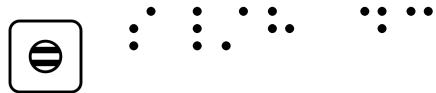
manual



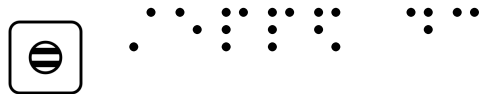
white



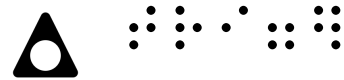
pink



s/h dc



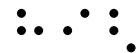
stepped dc



trigger

VCF

filter

 
high

 
band

 
low

 
Q

 
freq vc

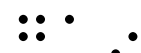
 
freq vc ×

 
high

 
band

 
low

 
input

 
gain

 
frequency

SSG

smooth & stepped generator



smooth vc



smooth vc x



smooth vc offset



stepped vc



stepped vc x



stepped vc offset



smooth out



smooth cycle



smooth hold



stepped sample



stepped cycle



stepped out



smooth in



coupler



coupler



stepped in

⠠⠠⠠⠠⠠ SEQ

⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ sequencer



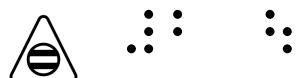
5 out



4 out



3 out



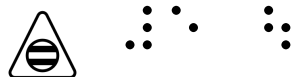
2 out



1 out



clock



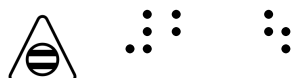
5 out



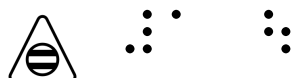
4 out



3 out



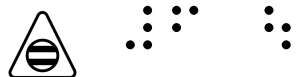
2 out



1 out



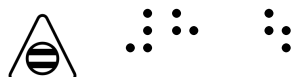
hold



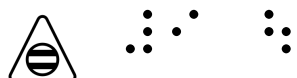
6 out



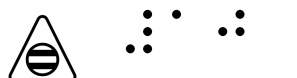
7 out



8 out



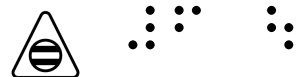
9 out



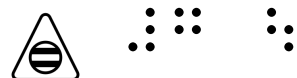
10 out



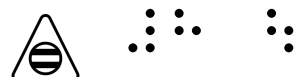
reset



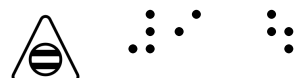
6 out



7 out



8 out



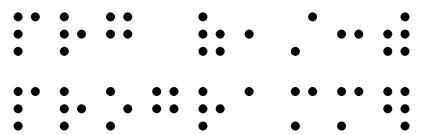
9 out



10 out



step



PRG (1/3) programmer



1 out



1 a



1 b



1 c



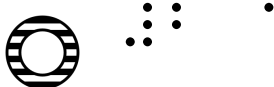
1 in



select



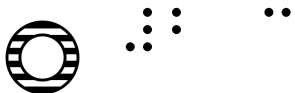
2 out



2 a



2 b



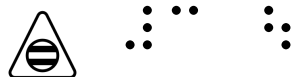
2 c



2 in



select



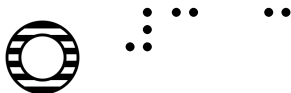
3 out



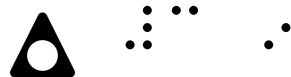
3 a



3 b



3 c



3 in



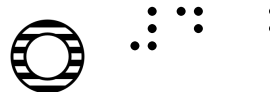
select



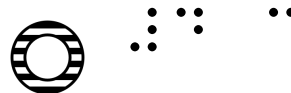
4 out



4 a



4 b



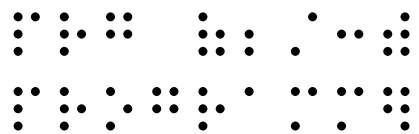
4 c



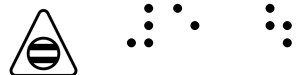
4 in



select



PRG (2/3) programmer



5 out



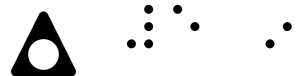
5 a



5 b



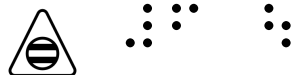
5 c



5 in



select



6 out



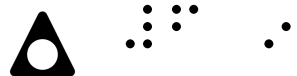
6 a



6 b



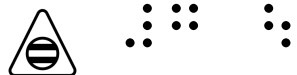
6 c



6 in



select



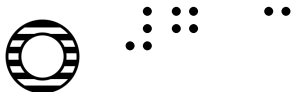
7 out



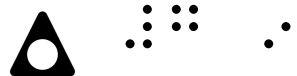
7 a



7 b



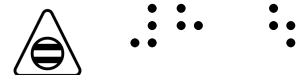
7 c



7 in



select



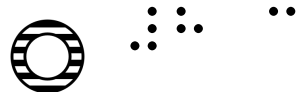
8 out



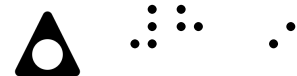
8 a



8 b



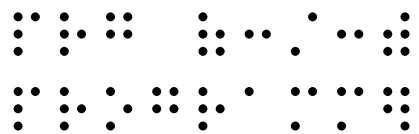
8 c



8 in



select



PRG (3/3) programmer



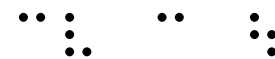
all out



cv a out



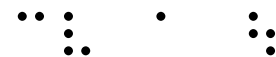
cv b out



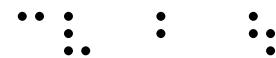
cv c out



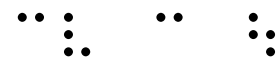
all out



cv a out



cv b out



cv c out

TBS

bi-directional router

